

1. Administrative & Study Identification

**Full Study Title:** A Phase 2, Randomized, Open-label Study of Relatlimab in Combination with Nivolumab in Participants with Advanced Hepatocellular Carcinoma who are Naive to IO Therapy but Progressed on Tyrosine Kinase Inhibitors **Short Title/Acronym:** RELATIVITY-073 **Protocol Number:** CA224-073 (EudraCT: 2018-003151-38) **NCT Number:** NCT04567615 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Relatlimab (LAG-3 inhibitor) and Nivolumab (PD-1 inhibitor) | **Drug Name:** nivolumab **Trade Names:** Opdualag, Opdivo **ATC Code:** L01FF01 **EMA Product Information URLs:** \* [https://www.ema.europa.eu/en/documents/product-information/opdivo-epar-product-information\\_en.pdf](https://www.ema.europa.eu/en/documents/product-information/opdivo-epar-product-information_en.pdf) \* [https://www.ema.europa.eu/en/documents/product-information/opdualag-epar-product-information\\_en.pdf](https://www.ema.europa.eu/en/documents/product-information/opdualag-epar-product-information_en.pdf) **Phase of Development:** Phase 2 (PHASE2) **Medical Monitor:** Bristol Myers Squibb Clinical Lead

---

2. Study Synopsis & Rationale

**2.1 Study Objective** \* **Primary Objective:** To evaluate the effectiveness, measured by Objective Response Rate (ORR) per RECIST v1.1, of relatlimab in combination with nivolumab in participants with advanced hepatocellular carcinoma (HCC). \* **Secondary Objectives:** To evaluate the Disease Control Rate (DCR), Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), and overall safety profile (adverse events).

**2.2 Background & Rationale** To address the need for effective second- and third-line therapies for advanced HCC patients who have progressed on tyrosine kinase inhibitors (TKIs). Because single-agent immune checkpoint inhibitors (ICIs) have limited response rates in HCC, dual checkpoint blockade targeting both the PD-1 (nivolumab) and LAG-3 (relatlimab) pathways aims to restore T-cell function, overcome immune evasion, and improve clinical outcomes in this immunotherapy-naive population.

---

3. Study Design & Methodology

**3.1 Design Framework** \* **Study Type:** Interventional \* **Control Type:** Active Comparator (Nivolumab monotherapy arm) \* **Blinding:** Open-label \* **Randomization:** Randomized (assigned to Nivolumab monotherapy or Relatlimab + Nivolumab combination regimens) \* **Trial Status:** ACTIVE\_NOT\_RECRUITING

**3.2 Planned Enrollment & Duration \* Target Enrollment (\$N\$):** 266 participants \* **Number of Sites:** Multicenter (Global) \* **Estimated Study Start:** 04-Feb-2021 \* **Estimated Primary Completion:** July 2023 (Approximately 29.5 months from randomization)

---

## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age  $\geq 18$  years with histological confirmation of advanced/metastatic hepatocellular carcinoma (HCC). 2. Immunotherapy (IO) treatment-naïve in the advanced/metastatic setting. 3. Have received 1 or 2 lines of tyrosine kinase inhibitor (TKI) therapies with evidence of radiographic progression on or after the last line of TKI therapy. 4. Child-Pugh score of 5 or 6 and Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 5. Must have at least one Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 measurable untreated lesion.

**4.2 Exclusion Criteria** 1. Known fibrolamellar HCC, sarcomatoid HCC, or combined hepatocellular cholangiocarcinoma. 2. Prior organ allograft or allogeneic bone marrow transplantation. 3. Uncontrolled or significant cardiovascular disease or active known autoimmune disease.

---

## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen \* Dosage/Frequency:** Intravenous (IV) administration of relatlimab + nivolumab combination or nivolumab alone, administered every 2 to 4 weeks (depending on the assigned cohort dosing schedule). \* **Route of Administration:** Intravenous (IV) \* **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent.

**5.2 Concomitant & Prohibited Therapy \* Permitted:** Standard palliative care, pain management, and symptom control. \* **Prohibited:** Concurrent anti-cancer therapies, systemic immunosuppressive medications (except for the management of immune-related adverse events), and live-attenuated vaccines.

---

## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days)    | Key Activities/Procedures                                                               |
|--------------|------------------|-----------------------------------------------------------------------------------------|
| Screening    | Day -28 to 0     | Informed Consent, Histology confirmation, Baseline Tumor Imaging, Labs                  |
| Baseline     | Day 0            | Randomization, First Dose Administration                                                |
| Interim      | Every 2-4 Weeks  | IV Infusion, Safety Labs, AE monitoring (Tumor imaging every 8 weeks)                   |
| End of Study | Post-Progression | Discontinuation visit, Safety follow-up (30-100 days post-last dose), Survival tracking |

---

## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint** \* **Definition:** Objective Response Rate (ORR) assessed by BICR. ORR is defined as the percentage of participants whose best overall response (BOR) is a confirmed Complete Response (CR) or confirmed Partial Response (PR) per RECIST v1.1 among all participants in the respective analysis set. Confirmation of response is required at least 4 weeks after the initial response. \* **Measurement Method:** Blinded Independent Central Review (BICR) of radiographic imaging.

**7.2 Secondary & Exploratory Endpoints** \* Disease Control Rate (DCR) assessed by BICR. \* Duration of Response (DOR) assessed by BICR. \* Progression-Free Survival (PFS) assessed by BICR. \* Overall Survival (OS) and incidence of treatment-related adverse events (TRAEs).

---

## 8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Assessed continuously and graded using CTCAE v5.0, with a specific focus on identifying and managing immune-related adverse events (irAEs).
  - **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness.
  - **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee oversees safety and efficacy data throughout the trial.
- 

## 9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for efficacy outcomes (all randomized patients); Safety Set for safety endpoints (all treated patients).
  - **Sample Size Justification:** Powered to detect a statistically significant improvement in the primary endpoint (ORR) across the treatment arms. Target  $N = 266$ .
  - **Handling of Missing Data:** Participants lacking post-baseline tumor assessments are treated as non-responders for ORR. Time-to-event endpoints (PFS, OS) are censored at the date of last contact or start of subsequent anti-cancer therapy.
- 

## 10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Routine on-site and remote monitoring visits to review eCRFs, source documents, and protocol compliance.

- **Audit Trail:** eCRF locking, query resolution processes, and potential audits by the sponsor Quality Assurance department or regulatory health authorities.
- 

## 11. Study Identifiers & Governance

**Primary ID:** CA224-073 **Registry Number:** NCT04567615 **Posting ID:** 604182846469196658 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study of Relatlimab in Combination With Nivolumab in Participants With Advanced Liver Cancer Who Have Never Been Treated With Immuno-oncology Therapy After Prior Treatment With Tyrosine Kinase Inhibitors **Official Regulatory Title:** A Phase 2, Randomized, Open-label Study of Relatlimab in Combination with Nivolumab in Participants with Advanced Hepatocellular Carcinoma who are Naïve to IO Therapy but Progressed on Tyrosine Kinase Inhibitors (RELATIVITY-073)

---

## 12. Clinical Framework (Hepatocellular Carcinoma Specific)

**Primary Condition (Preferred UMLS Name):** Hepatocellular Carcinoma (HCC) **Disease Areas:** Hepatocellular Carcinoma, Hepatoma, Liver Cancer, Adult, Liver Cell Carcinoma, Liver Cell Carcinoma, Adult **Diagnosis Threshold:** Advanced/metastatic HCC progressed on Tyrosine Kinase Inhibitors (Child-Pugh A [5-6]) **Medical Methodology:** Immuno-oncology combination therapy (Dual blockade of PD-1 and LAG-3 checkpoints) **Optimization Target:** Objective Response Rate (ORR) and extended Overall Survival (OS) **Target Population Ontology:** Adult (Preferred Name: Adult | Semantic Type: Age Group) \* **Definition:** An age group comprised of humans who have reached reproductive age. \* **MeSH Code:** D000328 \* **SNOMED ID:** 133936004 \* **SNOMED Hierarchy:** 410598002|Person categorized by age|Person categorized by age (person)

---

## 13. Study Procedures & Methodology

**Management Pathway:** Second/Third-line treatment pathway for advanced HCC **Education Protocol (HCP):** Site guidelines for IV infusion and management algorithms for immune-related adverse events (irAEs) **Education Protocol (Patient):** Education on recognizing and immediately reporting symptoms of immune toxicities **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for objective tumor response assessment

---

## 14. Observation & Follow-Up Schedule

**Total Duration:** Approximately 29.5 months to primary completion  
**Onsite Visit Cadence:** Every 2 to 4 weeks for infusion and

clinical assessment **Remote Monitoring Frequency:** Post-treatment survival follow-up every 12 weeks **Checklist Review Interval:** Radiographic tumor imaging every 8 weeks

---

**15. Sign-off & Approval**

|                                 |        |           |                        |      |      |             |      |
|---------------------------------|--------|-----------|------------------------|------|------|-------------|------|
| Role                            | Name   | Signature | Date                   | :--- | :--- | :---        | :--- |
| <b>Principal Investigator</b>   | [Name] | _____     | [DD- <i>MMM-YYYY</i> ] |      |      |             |      |
| <b>Clinical Operations Lead</b> | [Name] | _____     | [DD- <i>MMM-YYYY</i> ] |      |      | <i>Lead</i> |      |
| <i>Statistician</i>             | [Name] | _____     | [DD- <i>MMM-YYYY</i> ] |      |      |             |      |